MILITARY SPECIFICATION

BAG, URINE COLLECTION

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers three styles of disposable, plastic, closed system, sterile urine collection bags suitable for use in the collection and measurement of urinary output.

1.2 Classification.

1.2.1 Types.

Style A - Translucent with white opaque back

Style B - Translucent

Style C - Translucent with white opaque back and vented adapter.

2. APPLICABLE DOCUMENTS

2.1 Specification and Standards. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-585	Box, Wood, Wirebound.
PPP-B-601	Box, Wood, Cleated Plywood.
PPP-B-621	Box, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Shipping, Fiberboard.

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible.
MIL-S-36586	Sterilization Test Strip Set, Bacterial Spore.

FSC 6530

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of Military specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposal shall apply.

UNITED STATES PHARMACOPEIAL CONVENTION, INC. (U.S.P.)
Pharmacopeia of the United States

(Applications for copies should be addressed to the Mack Publishing Company, Easton, Pennsylvania 18042.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
D882 Method of Test for Tensile Properties of Thin Plastic Sheeting

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Material. The material shall be as specified herein, of the highest quality throughout and suitable in all respects for the purpose intended.

3.1.1 Polyvinyl chloride (PVC). The PVC sheets shall be formulated from virgin resin, and shall have a minimum tensile strength of 2000 p.s.i. and a minimum elongation of 200 percent when tested as specified in 4.4.2. Where white pigment has been added to the PVC, an average elongation of 175 percent shall be acceptable.

3.1.2 Wire. The wire hangers, if applicable, shall be 0.120 inch, (No. 11 American Steel Wire gage) cold drawn steel, zinc, cadmium or tin plated.

3.1.3 Cord. The cord used with the wire hangers shall be nylon, cotton, or wax impregnated cotton, tightly woven, 20 to 36 inches long and 3/32 inch minimum diameter. The cord shall be furnished with a suitable corrosion-resistant hook, capable of supporting the weight of a bag filled with 2000 cc, of water.

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3.2 Construction. The sterile urine collection bag, shall consist of a urine collecting bag, an inlet tube with a safety chamber, a bottom drain tube with a shut-off clamp, and a hanger for suspension from the bed, chair or patient. The closed system shall be either styles A, B or C, in accordance with Figure 1 for Style A or Figure 2 for Style B or Figure 3 for Style C. The collection bag shall be packaged in a polyethylene bag or polyethylene/polymylar pouch to maintain sterility of the bag until manually opened. There shall be no water inside the plastic overwrapper or the collection bag when tested as specified in 4.4.3. The collection bag shall be sufficiently transparent or translucent to observe the flow and contents. The bag shall be capable of suspension by the catheter adapter, and no joint or connection shall either separate, or begin to separate, when tested as specified in 4.4.4.

3.2.1 Bag. The urine collection bag shall be constructed of two sheets of PVC, heat sealed on all edges and vented at the top or in the catheter adapter. The bag shall not leak water when tested as specified in 4.4.5. The bag shall not burst at a pressure less than 150 mm Hg when tested as specified in 4.4.8. The bag shall have a capacity of at least 2000 cubic centimeters (cc). It shall be fitted with a horizontal air vent at the top, kept open with a rigid tube in the channel. As an alternate, a circular filter vent secured over a port in the top portion of the bag or an air vent incorporated in the catheter shall be acceptable.

3.2.1.1 Graduations and markings. Graduations and markings shall be clear and indelible when tested as specified in 4.4.6. They shall not be attached or secured with adhesives. The numbers and letters shall be a minimum of 1/8 inch in height.

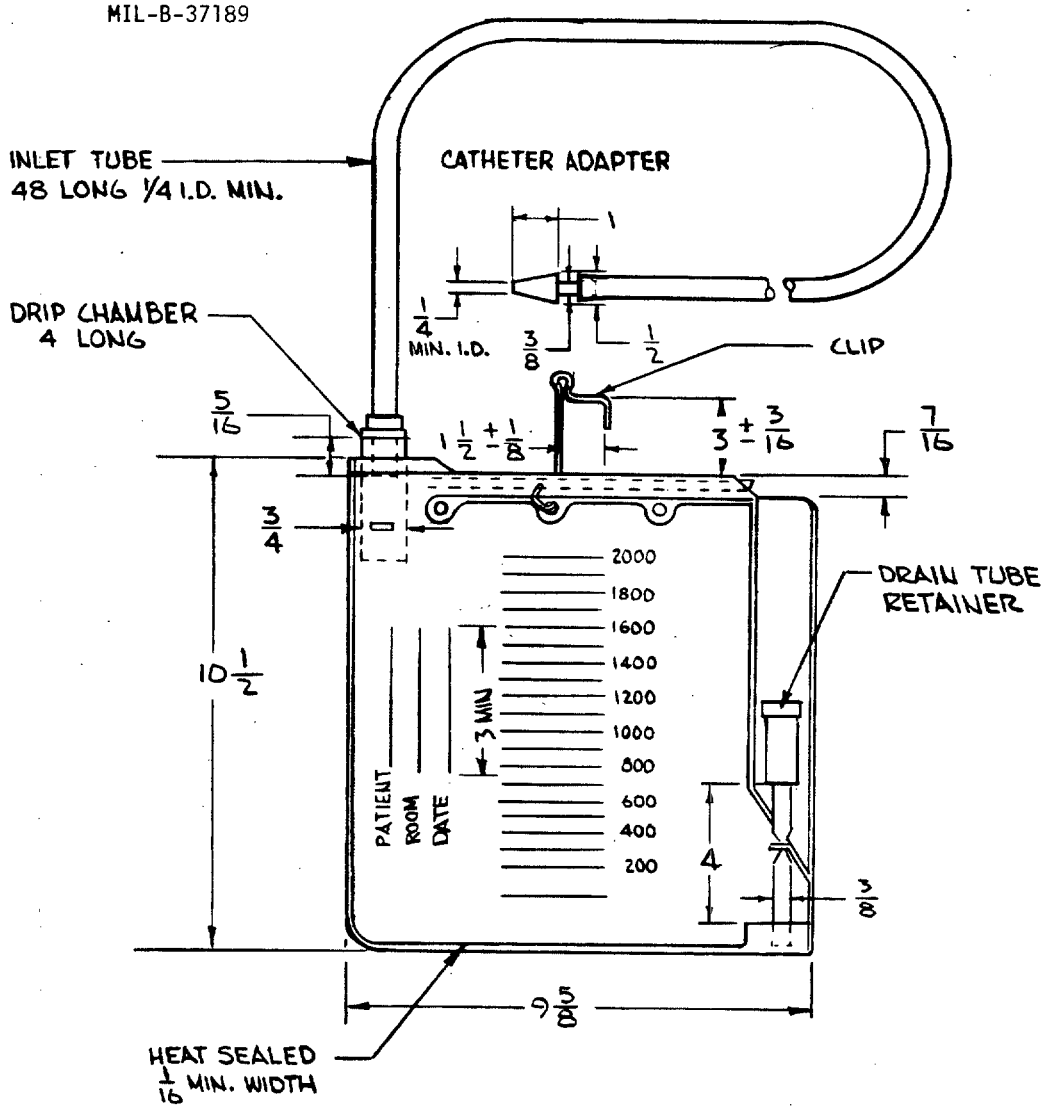
3.2.1.1.1 Graduations. Graduations shall be in 100 cc steps beginning at 100 cc and ending at not less than 2000 cc but not more than 2500 cc. At least every other graduation, beginning at 200 cc, shall be identified. Additional graduations shall be permissible in a lower corner.

3.2.1.1.2 Marking. Other marking shall be as shown on Figures 1, 2 or 3. The manufacturer's name or registered trademark shall be printed on the Bag.

3.2.1.2 Volumetric accuracy. The bag shall have a volumetric accuracy of $\pm$ 10 percent when tested as specified in 4.4.7.

3.2.2 Safety chamber. The safety chamber shall be of rigid plastic and shall be permanently attached to the inlet tube and collection bag. As an alternate, the vent may be located in the catheter adapter. The design shall be such as to prevent the retrograde migration of bacteria.

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TOLERANCES

UP TO 1 INCH ± 1/16
1 UP TO 6 INCH ± 1/8
6 UP TO 12 INCH ± 1/4
12 OR GREATER ± 1/2

BAG - PVC SHEETS 0.0085 MINIMUM THICKNESS

ALL DIMENSIONS ARE IN INCHES.

FIGURE 1 - STYLE A - BAG, URINE COLLECTION

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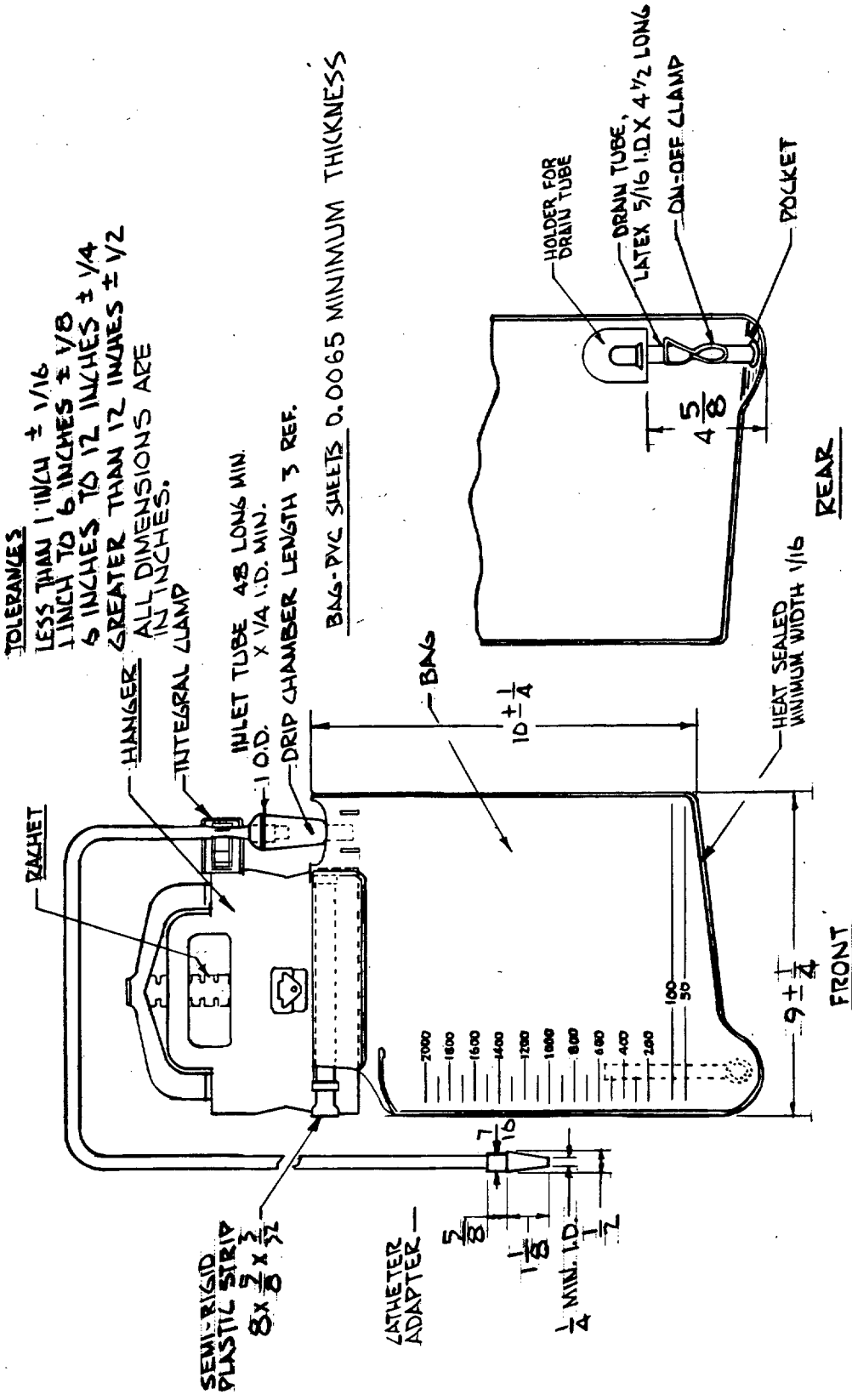


FIGURE 2- STYLE B- BAG, URINE COLLECTION

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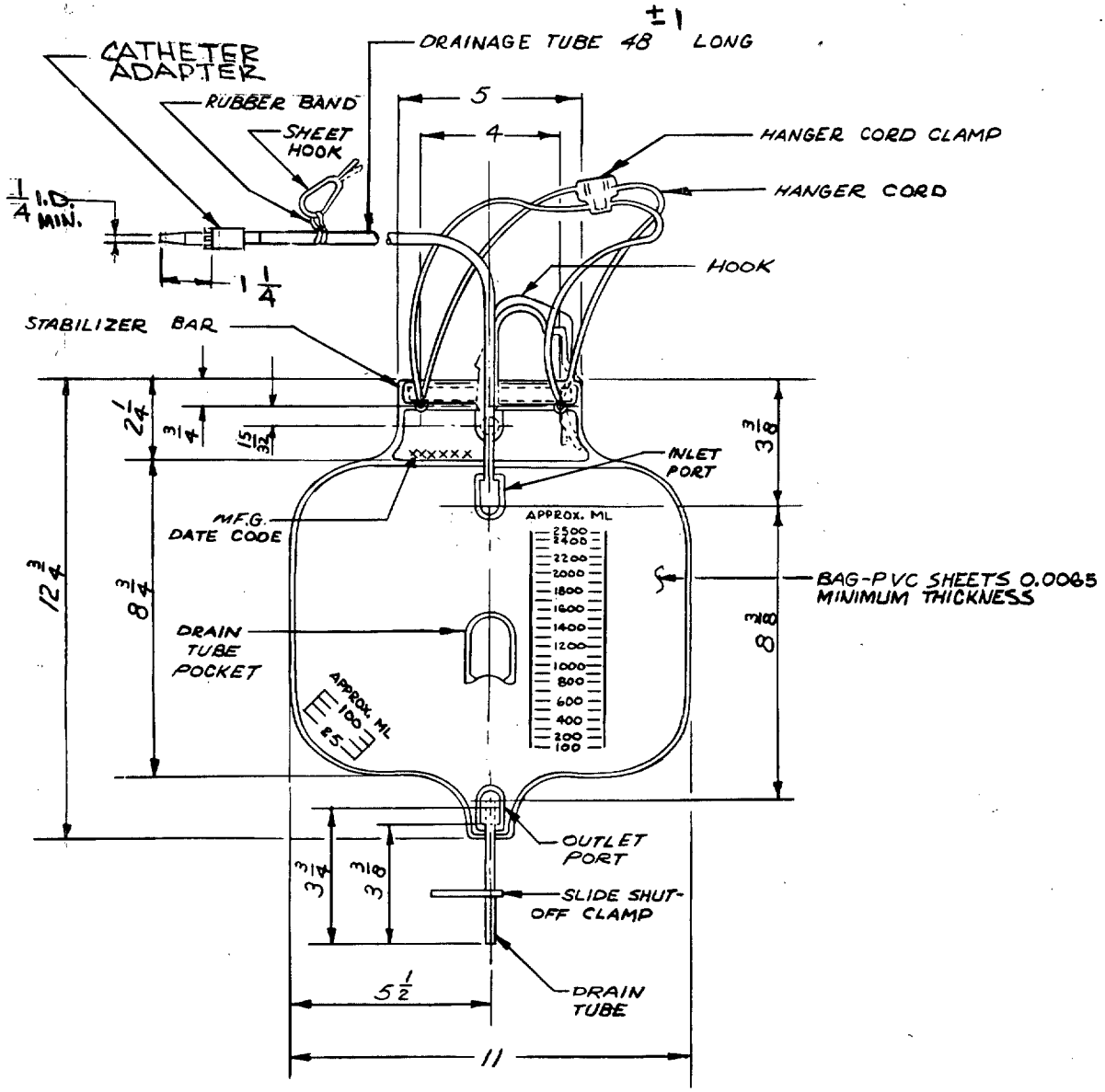


FIGURE 3-STYLE C-BAG, URINE COLLECTION

NOTES:

1. TOLERANCES; UP TO 1" ±1/16, 1" UP TO 6" ±1/8, 6" UP TO 12" ±1/4, 12" OR GREATER ±1/2" UNLESS OTHERWISE SPECIFIED
2. HEAT SEALS 1/16" MINIMUM WIDTH.
3. ALL DIMENSIONS ARE IN INCHES.

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3.2.3 Inlet tube. The inlet tube shall have a minimum wall thickness of 0.040 inch. The PVC inlet tube shall be permanently attached to the catheter adapter and the safety chamber or inlet port. The catheter adapter shall be fitted with a tip protector.

3.2.4 Drain. The PVC or latex drain shall be permanently attached to the bottom of the bag. To prevent drip, the end of the drain tube shall be placed in a permanent pocket or socket, located directly above the drain. The drain tube shall be furnished with an ON - OFF clamp to seal the drain tube. The clamp shall be of a design and manufacture to avoid danger of puncturing the bag.

3.2.5 Sterility. The contents of the plastic overwrapper shall be sterile when tested as specified in 4.4.1.

3.2.5.1 Certificate of sterility. A certificate of sterility, stating that samples from each sterilizer lot have been tested and found to be sterile, shall be submitted to the procuring activity. The certificate of sterility shall include:

a. Item identification.
b. National stock number.
c. Contract or purchase order number.
d. Sterility lot or control number.
e. Date of sterilization.
f. Quantity of product in sterilizer lot.
g. Type of sterilization (gas, radiation, heat, or steam).
h. Form of biological indicator (i.e., inoculated paper spore strip or carrier, inoculated product, or inoculated simulated product).
i. Test organism(s) and population density used.
j. Test medium, incubation temperature, incubation time, and dates of tests.
k. Number of test units.
l. Statement that samples comply with sterility tests.
m. Signature of head of laboratory or a designated qualified alternate.

When inspection is made at destination, the certificate of sterility shall be submitted by the contractor to the Directorate of Medical Materiel, ATTN: DPSC-ATQ, Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, PA 19101, and a copy forwarded with each shipment to the consignee. When inspection is made at source, certificate of sterility shall be furnished to the cognizant Government Quality Assurance Representative for submission to the procuring activity.

3.2.5.2 Sterility lot. A sterility lot is that single quantity of product subjected to the same manufacturing operation and simultaneously sterilized in the same sterilizer chamber. Each unit package shall be clearly marked with a sterility lot (control) number and date of sterilization as specified in 5.3.1. Each sterility lot shall be traceable to the pertinent manufacturing lot(s).

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3.2.6 Instructions. Each individual unit shall be furnished with instructions for suspending, using and emptying of the system.

3.3 Workmanship. Workmanship shall be first class throughout. The urine collection bags shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the components may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 Unit of product. The unit of product, for inspection and test purposes only, shall be one urine collection bag.

4.2.2 For examination. Three units from each production lot shall be examined for dimensional defects. The inspection level and acceptable quality level (AQL) shall be as indicated in Table I and in accordance with MIL-STD-105.

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TABLE I

Component	Type of Defect	Inspection Level Unit of Product	AQL (percent defective)*
Collection Bag	Major defects	S-3	1.0
	Minor defects	S-3	4.0

* If acceptance number is not exceeded, lot shall be accepted.

4.2.3 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and Table II.

TABLE II

	Characteristic	Req't	Test Procedure	Inspection Level	Unit of Product
Component					
PVC Sheeting	Tensile strength	3.1.1	4.4.2	**	*
	Elongation	3.1.1	4.4.2	**	*
End Item (see 4.2.4)					
	Submersion	3.2	4.4.3	S-2	2.5
	Pull	3.2	4.4.4	S-1	2.5
	Leakage	3.2.1	4.4.5	S-3	1.0
	Indelible	3.2.1.1	4.4.6	S-1	2.5
	Volume	3.2.1.2	4.4.7	S-1	2.5
	Burst	3.2.1	4.4.8	S-1	2.5
	Sterility	3.2.5	4.4.1	***S-1	*

* Acceptance number shall be zero for all sample sizes.
** Each lot of sheeting shall be tested.
*** Sample size shall not be less than the minimum specified in 4.4.1.

4.2.4 Test samples. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

4.3 Examination. The collection system shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the classification of defects listed in Table III.

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TABLE III

Categories	Defects*
Major	
101	Bag not free from stains, cracks, other imperfections.
102	Bag not properly sealed.
103	A component not sealed.
104	Heat seals not minimum width.
105	Plastic overwrapper shall be free from cuts, tears, etc.
Minor	
201	Drain control valve not capable of sealing without damaging the tube.
202	Marking on bag not clear.
203	Instructions for use not clear or complete.

* Examination shall not be restricted to the classified possible defects listed above.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Sterility test. The contents of the plastic overwrapper shall be tested for sterility in accordance with the method specified in the U.S.P. for nonparenteral devices using inoculated product, inoculated simulated product, or paper spore strip (carrier). Product containing a biological indicator shall be marked or labeled so that it clearly differentiates from product intended for distribution. The incubation medium and minimum time in each case shall be soybean casein digest for 7 days for inoculated product, inoculated simulated product, or paper spore stirp (carrier). The incubation temperature shall be as indicated in paragraph 4.4.1.1. The minimum number of test units, regardless of lot size, shall be 10 except for inoculated carrier which shall be 20 for Type I Sterilization and 30 for Type II Sterilization including inoculated carrier and product test units. The inspection level shall be as indicated in Table II.

4.4.1.1 Inoculated product or simulated product. For each type of sterilization, the organism used shall be as listed below:

Type of Sterilization	Organism	Incubation Temperature
Steam	B. stearothermophilus	55° - 65°C
Dry Heat	B. stearothermophilus, or	55° - 65°C
	B. subtilis var. niger (or globigii)	30° - 35°C
Gas	B. subtilis var. niger (or globigii)	30° - 35°C
Radiation	B. pumilus	30° - 35°C

The number of viable cells shall be sufficient to provide assurance that the product has been sterilized.

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4.4.1.2 Inoculated carrier. Inoculated carriers (paper spore strips) if used shall be in accordance with MIL-S-36586. Other paper carriers shall be inoculated to be equivalent resistance. The method of incubation and number of test units shall be as indicated, inoculated product or simulated product. The method of sterilization will control the organism in the spore strip to be tested for as shown in 4.4.1.1. Type B, Bacillus subtilis, var. globigii, as indicated in MIL-S-36586, shall be used for testing the subject bag. NOTE: When an inoculated carrier is chosen, product samples must also be tested as required by the U.S.P.

4.4.1.3 Alternate method. Suppliers conducting sterility tests or designating lots in a manner other than that indicated herein shall submit proof that the alternate method provides equal assurance concerning the sterility of the product. However, all sterility tests shall be conducted using biological indicators. Requests for approval of any alternate method of testing for sterility shall be made to the Contracting Officer, and approval obtained before employment of the alternate method on a government contract.

4.4.1.4 Product testing. Testing for sterility of the product after receipt by the government shall be without a biological indicator and in accordance with the U.S.P.

4.4.2 Tensile strength and elongation. The plastic sheeting shall be tested for tensile strength and elongation in accordance with ASTM D882 Method of Test for Tensile Properties of Thin Plastic Sheeting.

4.4.3 Submersion test. The complete bag in its plastic overwrapper shall be submerged in hot water (90°F.), at least one-inch below the water surface. Observation of evolution of air bubbles shall be made at each postion of the sample. Bubbles which appear on the surface of the package but are not release or are released at a slowly decreasing rate are not to be construed as indication of failure. A steady stream or recurring succession of bubbles from any surface or seam shall be cause for rejection.

4.4.4 Pull test. The complete system shall be filled to the 2000 cc level and suspended by the catheter adapter for one-hour. The system shall be checked at each joint or connection for separation.

4.4.5 Leakage test. The urine collection bag shall be filled with 2000 cc of water at room temperature and suspended for 96 hours. The bag shall then be inspected for leakage.

4.4.6 Indelible test. Markings shall not fade or disappear when rubbed 10 complete cycles (forward and back) with a water moistened cloth.

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4.4.7 Volume test. The urine collection bag shall be placed in its normal operating position in accordance with the manufacturer's instructions. Water at room temperature shall be added. The accuracy at 1000 cc, 1600 cc and 2000 cc shall be determined.

4.4.8 Burst test. The urine collection bag shall be inflated with compressed air through the connecting tube. Prior to inflation, any air vents or filters shall be sealed with an adhesive to prevent loss of air. Internal pressure shall be stabilized at 20 mm Hg and then increased through a regulator valve at a rate of 1 mm Hq per second. The pressure at which a leak or burst occurs shall be recorded.

4.5 Examination for preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of Section 5. Defects shall be scored as specified in Table IV. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 defects per hundred units.

Table IV. Examination for Preparation for Delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Packaging and packing materials	Any component missing or damaged.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, tape improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper size, location, sequence, or method of application.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Unit of issue. One package containing twenty (20) bags, as specified, constitutes one unit of issue.

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5.1.1.1 Level A.

5.1.1.1.1 Individual package. Each collection bag shall be packaged in a sealed peel type bag of appropriate size capable of maintaining sterility of contents unless damaged or opened. Metal components of the item shall be protected in such a manner so as to prevent damage to the item or polyethylene bag (individual package).

5.1.1.1.2 Unit package. Unit package shall be a snug fitting box constructed in accordance with PPP-B-636 type CF, class domestic. Closure shall be adequate to prevent accidental opening.

5.1.1.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be Level A, B or C as specified (see 6.2).

5.2.1 Level A. Two (2) units of issue shall be packed in a box designed for a type 2 load, and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2 or PPP-B-636 type CF, class weather-resistant. Closure and strapping shall be in accordance with the appendix of the applicable box specification.

5.2.1.1 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard boxes shall be waterproofed in accordance with PPP-B-636.

5.2.2 Level B. Two (2) units of issue shall be packed in a box designed for a type 2 load, and constructed in accordance with PPP-B-585, class 2, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, type CF, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Individual package. Each package shall be commercially identified. Sterility lot (control) number and the following or similar legends shall be shown: "STERILE" and "STERILITY GUARANTEED UNLESS PACKAGE IS DAMAGED OR OPENED."

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5.3.2 Unit package. Each unit package shall be marked in accordance with Military Standard MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Sterility lot (control) number, contract or purchase order number and name of contractor shall be shown. The method of sterilization (G-Gas, R-Radiation, C-Cold, S-Steam, D-Dry Heat Sterilization) shall be shown following the level of packaging and the date of packaging. Example: "A-1/75-G". Final sterilization shall be on or after indicated packaging date.

5.3.3 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. Sterility lot control number shall be shown. The method of sterilization as specified in 5.3.2 shall also be shown.

6. NOTES

6.1 Intended use. Shall be suitable for the collection and measurement of urinary output without disagreeable odor or external contamination.

6.2 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 This specification covers three styles of the following item appearing in the Federal Supply Catalog:

National Stock Number	Item Identification
6530-00-105-8649	BAG, URINE COLLECTION, Plastic, Disposable, 20s

6.4 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DSA-DM

Project No. 6530-1332
cs

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-B-37189 BAG, URINE COLLECTION

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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